PARENT–TEACHER COMMUNICATION SYSTEM

School Management System

Technical Specification & Implementation Guide — v1.0 · March 2026

Confidential · Internal Documentation

Audience
Development Team · Product Team · School Administrators

Related Modules
Parent Module · Student Module · Staff Module · Notification System · Calendar & Events · Security

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1. MODULE OVERVIEW

The **Parent–Teacher Communication System** enables secure, structured communication between guardians and teaching staff while preventing misuse or excessive messaging.

The system is designed to:

- encourage **constructive academic discussions**
- prevent **communication overload for teachers**
- maintain **auditability and transparency**
- ensure **student privacy and discipline governance**

The system supports **messaging, meeting scheduling, announcements, and academic discussions**.

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2. CORE SUBMODULES

Submodule	Description	Priority	Status
Messaging System	Parent ↔ Teacher conversation threads	Core	Required
Meeting Scheduler	Parent-teacher meeting requests	Core	Required
Teacher Office Hours	Teacher availability management	Core	Required
Notification Engine	Alerts for new messages and meetings	Core	Required
Moderation & Escalation	Admin oversight and conflict escalation	Enhanced	Recommended
Communication Logs	Message history and audit trails	Core	Required

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3. COMMUNICATION DESIGN PRINCIPLES

The system enforces **structured communication instead of open chat**.

This ensures:

```
Respect for teachers' time
Professional conversations
Traceable discussions
Controlled escalation
```

Parents can communicate **only with teachers responsible for their child**.

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4. ROLE-BASED COMMUNICATION ACCESS

Role	Communication Scope
Parent	Teachers assigned to their child
Teacher	Students they teach
Admin	Full oversight
Student	Limited messaging (optional feature)

Parents cannot message:

- random teachers
- school administrators directly
- teachers outside their child's subjects

5. MESSAGING SYSTEM

Conversation Model

Communication is organized by **student + subject + teacher**.

Example thread:

```
Student: John Doe
Subject: Mathematics
Teacher: Mr. Smith

Conversation Thread
```

Messaging Features

Feature	Description
Threaded conversations	Organized by student and subject
Attachments	Parents may send documents
Message status	Sent / Delivered / Read
Conversation history	Stored permanently

Message Structure

```
Message
│
├ sender_id
├ receiver_id
├ student_id
├ subject_id
├ message_body
├ attachments
├ timestamp
└ status
```

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6. MESSAGE RULES & LIMITATIONS

To prevent excessive messaging, the system enforces limits.

Rule	Description
Message hours	7 AM – 7 PM
Daily message limit	5 messages per parent
Teacher response expectation	24–48 hours

Urgent matters must be directed to **school administration**.

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7. TEACHER OFFICE HOURS

Teachers define availability windows for communication.

Example:

```
Monday: 3PM – 5PM
Wednesday: 4PM – 6PM
Friday: 2PM – 4PM
```

Parents can send messages anytime but **teachers receive priority notifications during office hours**.

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8. PARENT–TEACHER MEETING SCHEDULER

The system includes a **formal meeting request system**.

Meeting Request Workflow

```
Parent
   │
   └ Request Meeting
            │
            ├ Choose Teacher
            ├ Select Student
            ├ Select Time Slot
            └ Add Meeting Topic
```

Teacher then:

```
Accept
Decline
Suggest new time
```

Meeting Types

Type	Description
Virtual	Online meeting
Physical	On-campus meeting
School Event	Scheduled PTM

Virtual meetings may integrate with:

- Zoom
- Google Meet
- Microsoft Teams

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9. SCHOOL ANNOUNCEMENTS

Teachers may broadcast announcements to parents.

Examples:

- class project updates
- exam reminders
- field trip notices

Announcements appear on:

```
Parent Dashboard
Notification Center
Email/SMS
```

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10. NOTIFICATION SYSTEM

Notifications ensure timely communication.

Notification Triggers

Event	Notification
New message	Parent/Teacher alert
Meeting request	Teacher alert
Meeting approval	Parent alert
Announcement	Parent alert

Delivery channels include:

```
In-app notification
Email
SMS
```

SMS may integrate with systems such as:

- Twilio
- Africa's Talking

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11. MODERATION & ESCALATION

The system must support **administrative oversight**.

Escalation Workflow

If communication becomes inappropriate:

```
Parent Complaint
        |
Teacher Escalates
        |
Admin Review
        |
Resolution
```

Administrators can:

- review message history
- intervene in conversations
- restrict accounts

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12. COMMUNICATION LOGGING

Every interaction must be logged for governance and compliance.

Logged Events

Event
message_sent
message_read
meeting_requested
meeting_accepted
meeting_cancelled

Log fields include:

Field	Description
user_id	sender
action	system event
ip_address	device location
timestamp	event time

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13. DATA PRIVACY RULES

Student privacy must be protected.

Parents can only access:

```
their own child’s discussions
```

Teachers cannot share:

- confidential student information
- disciplinary records
- other student information

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14. API ENDPOINT STRUCTURE

Communication endpoints should use:

```
/api/communication/
```

Core APIs

Method	Endpoint	Description
GET	/api/communication/threads	List conversations
POST	/api/communication/message	Send message
GET	/api/communication/messages/{thread_id}	Message history
POST	/api/communication/meeting/request	Create meeting
GET	/api/communication/meeting/list	View meetings

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15. DATABASE TABLE STRUCTURE

Messages Table

Field	Type
id	UUID
sender_id	FK
receiver_id	FK
student_id	FK
subject_id	FK
message_text	Text
attachment_url	String
status	Enum
timestamp	Datetime

Meetings Table

Field	Type
id	UUID
parent_id	FK
teacher_id	FK
student_id	FK
meeting_time	Datetime
meeting_type	Enum
status	Enum

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16. INTEGRATION POINTS

Module	Purpose
Parent Module	Guardian access
Student Module	Student context
Staff Module	Teacher accounts
Calendar Module	Scheduling meetings
Notification System	Alerts
Security Module	Access control

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17. IMPLEMENTATION WORKFLOW

Communication lifecycle:

```
Parent opens dashboard
        |
Select child
        |
Select subject
        |
Open teacher conversation
        |
Send message or request meeting
        |
Teacher responds
        |
Conversation stored in communication log
```

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18. GOVERNANCE & CONTROL

To maintain professionalism:

- communication logs cannot be deleted
- abusive messaging can be flagged
- administrators have full oversight

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19. FUTURE ENHANCEMENTS

Future platform improvements may include:

- AI message moderation
- automatic meeting reminders
- translation for multilingual parents
- parent satisfaction analytics

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20. DESIGN PRINCIPLE

The Parent–Teacher Communication System follows three principles:

```
Structured communication
Teacher workload protection
Parent engagement
```

This ensures the platform supports **healthy school–family collaboration** while maintaining operational control.